

HR & Attendance Management System (HRM)

System Architecture, Modules, Tech Stack & Workflow Guide

Enterprise-Grade Technical & Functional Documentation

1. Executive Overview

The HR & Attendance Management System (HRM) is a modern web application designed to streamline company-wide attendance tracking, automated leave and permission management, employee milestone celebrations, GPS check-ins, and dynamic payroll calculations.

2. Technology Stack & Languages Used

- **Backend Framework:** PHP 8.2+ powered by Laravel 11/12 (MVC Architecture, Eloquent ORM, REST APIs, Migrations & Seeders).
- **Frontend & UI:** Laravel Blade Engine, HTML5, CSS3, Tailwind CSS, Bootstrap 5 UI & Bootstrap Icons.
- **Client Scripting:** JavaScript (ES6+), jQuery, AJAX for asynchronous instant updates without full page reloads.
- **Database Engine:** MySQL / MariaDB (Production) & SQLite (Testing / Development).
- **APIs & Libraries:** OpenStreetMap Nominatim API (Reverse GPS geocoding), PhpSpreadsheet (Excel data imports), Carbon (Date/Time math), DomPDF & PDFKit.

3. Core Functional Modules

Module A: Smart Attendance & Geolocation Check-In / Out

- **One-Click Check-In/Out:** Employees clock in/out with automatic GPS coordinate capture and street address resolution.
- **Status Evaluation:** ≥ 8 hours = Present (Full Day), 5 to 8 hours = Half Day, < 5 hours = Absent.
- **Auto Clock-Out at 18:30:** Automatically closes forgotten check-ins at 6:30 PM with precise working hour status.
- **Daily Work Reports:** Employees submit a concise task summary log upon clocking out.

Module B: Leave & Permission Management Policy

- **Request Category Selector:** Interactive UI toggle between Standard Leave and 1-Hour Permission.
- **1-Hour Permission Auto-Approval:** Requests up to 2 permissions (2 hours total) per calendar month are auto-approved instantly.
- **Automatic Quota Disabling:** Once 2 permissions are used in a month, the permission button is automatically disabled ("Disabled - 2/2 Used"), prompting the employee to apply for leave.
- **Leave Balance Tracking:** Real-time counters for Allocated Leaves, Used Leaves, Remaining Balance, and Pending Requests.

Module C: Automated Employee Celebrations & Notifications

- **Work Anniversary Celebrations:** Automatically delivers congratulatory messages on employee join_date anniversary (≥ 1 year completed).
- **Birthday Greetings:** Delivers warm celebratory birthday wishes on each employee's date of birth (dob).
- **Celebration Banners:** Vibrant congratulatory banner displayed at the top of the employee dashboard on their special day.
- **Duplicate Prevention:** Strict safeguards guarantee celebration messages fire exactly once per employee per calendar year.
- **Admin Notifications:** Real-time alerts inform HR administrators of milestone anniversaries, birthdays, and leave applications.

Module D: Dynamic Salary & Payroll Calculations

- **Payroll Formula:** $\text{Net Salary} = (\text{Base Salary} / 30) * \text{Worked Days}$ (where $\text{Worked Days} = \text{Present} + 0.5 * \text{HalfDay} + \text{Approved Paid Leaves}$).
- **Join-Date Proration:** Automatically calculates eligible days for new joiners starting mid-month.
- **Records & Preview:** Provides monthly salary previews and historical calculation records.

Module E: Multi-Branch & Administrative Control

- **Branch Scoping:** Branch Managers only manage employees in their branch, while Super Admins manage all branches.
- **Employee Directory:** Complete employee profiles with photo, designation, emergency contacts, and blood group.
- **Holiday Calendar & Old Data:** Holiday management and bulk spreadsheet Excel data import/export.

4. System Workflow & Process Lifecycle

A. Employee Daily Journey

1. Login: Employee logs in with their Employee ID at /employee/login.
2. Check-In: Hits Clock In; browser captures GPS coordinates and resolves location address.
3. Leave / Permission: Selects Leave (Sick, Casual, Earned, Half Day) or Permission (1 hr auto-approved, max 2/month).
4. Check-Out: Submits daily work summary; system calculates hours and finalizes attendance status.

B. Admin Management Journey

1. Dashboard Monitoring: Real-time overview of staff attendance, leaves, and pending approvals.
2. Approvals: Reviews and acts on pending leaves with automated attendance updating.
3. Payroll Generation: Runs salary preview and calculation at month end.

C. Background Automation Lifecycle

1. 18:30:00 Daily: Unclosed attendances are automatically closed with calculated status.
2. Daily Check: Scans join_date and dob across active employees to generate anniversary and birthday notifications.
3. 23:59:00 Daily: Non-attending active employees are automatically marked absent.

5. Installation & Quick Start

1. Install dependencies: `composer install && npm install && npm run build`
2. Setup Environment: `cp .env.example .env && php artisan key:generate`
3. Run Database: `php artisan migrate --seed`
4. Start Server: `php artisan serve` (Access at `http://127.0.0.1:8000`)
5. Run Automated Tests: `php artisan test`